



Optimizing dalbavancin dosing for complicated MRSA infections: a multicenter population pharmacokinetic study

Manjunath P. Pai, Pier Giorgio Cojutti, Riccardo De Paola, Eleonora Zamparini, Sara Tedeschi, Simone Giuliano, Marianna Meschiari, Alessia Prezzavento, Lorenzo Zammarchi, Pierluigi Viale, Federico Pea

PII: S1198-743X(26)00312-5

DOI: <https://doi.org/10.1016/j.cmi.2026.05.049>

Reference: CMI 4458

To appear in: *Clinical Microbiology and Infection*

Received Date: 4 March 2026

Revised Date: 20 May 2026

Accepted Date: 30 May 2026

Please cite this article as: Pai MP, Cojutti PG, De Paola R, Zamparini E, Tedeschi S, Giuliano S, Meschiari M, Prezzavento A, Zammarchi L, Viale P, Pea F, Optimizing dalbavancin dosing for complicated MRSA infections: a multicenter population pharmacokinetic study, *Clinical Microbiology and Infection*, <https://doi.org/10.1016/j.cmi.2026.05.049>.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Optimizing dalbavancin dosing for complicated MRSA infections: a multicenter population pharmacokinetic study

Manjunath P. Pai ^{1,*}, Pier Giorgio Cojutti ^{2,3}, Riccardo De Paola ^{2,3}, Eleonora Zamparini ⁴, Sara Tedeschi ^{2,4}, Simone Giuliano ⁵, Marianna Meschiari ⁶, Alessia Prezzavento ⁷, Lorenzo Zammarchi ^{8,9}, Pierluigi Viale ^{2,4}, Federico Pea ^{2,3}

¹ Department of Clinical Pharmacy, College of Pharmacy, University of Michigan, Ann Arbor, MI, USA

² Department of Medical and Surgical Sciences, Alma Mater Studiorum, University of Bologna, Bologna, Italy

³ Clinical Pharmacology Unit, IRCCS Azienda Ospedaliero-Universitaria di Bologna, Bologna, Italy

⁴ Infectious Diseases Unit, IRCCS Azienda Ospedaliero-Universitaria di Bologna, Bologna, Italy

⁵ Infectious Diseases Clinic, Santa Maria della Misericordia University Hospital of Udine, ASUFC, Udine, Italy.

⁶ Department of Surgery, Medicine, Dentistry and Morphological Sciences, University of Modena and Reggio Emilia, Modena, Italy

⁷ Unit of Infectious Diseases, Santa Chiara Hospital, APSS, Trento, Italy

⁸ Department of Experimental and Clinical Medicine, University of Florence, Florence, Italy

⁹ Department of Infectious and Tropical Diseases, Azienda Ospedaliero Universitaria Careggi, Florence, Italy.

***Corresponding author:** Manjunath (Amit) P. Pai, PharmD, FCP. Professor and Chair of Clinical Pharmacy, College of Pharmacy, University of Michigan-Ann Arbor, 428 Church St., Ann Arbor, MI 48108

Word Count: 2682/2700 words

Abstract: 293/300 words

References: 27/40 references

ABSTRACT

Objectives. Dalbavancin is increasingly used off-label for complicated methicillin-resistant *Staphylococcus aureus* (MRSA) infections requiring prolonged antimicrobial exposure. However, approved dosing regimens were developed for short-course therapy and may provide inadequate exposure in patients with obesity or preserved kidney

function. We aimed to develop a practical dosing strategy for prolonged dalbavancin therapy using population pharmacokinetic modeling and simulation.

Methods. We conducted a multicenter cohort study of adults treated with dalbavancin for presumed or documented MRSA infections across five centers. Dalbavancin plasma concentrations were measured using a validated LC–MS/MS assay. Population pharmacokinetic modeling was performed using nonlinear mixed-effects methods with evaluation of body size and kidney function covariates. Monte Carlo simulations (n=1000 per scenario) estimated the probability of target attainment (PTA) for maintaining dalbavancin concentrations ≥ 8.04 mg/L through 4 weeks across body mass index (BMI) and estimated glomerular filtration rate (eGFR) strata. Candidate regimens for 4–8 weeks of coverage were evaluated, with protein-binding sensitivity analyses across 93%–99%.

Results. A total of 311 patients contributed 946 dalbavancin concentrations with follow-up extending beyond 8 weeks. Compared with normal-weight patients, those with obesity (n=64) exhibited 49% higher clearance and 81% higher peripheral volume of distribution. Standard labeled regimens achieved 4-week PTA in only 50–60% of obese patients, with the lowest attainment observed among those with eGFR ≥ 120 mL/min. BMI- and eGFR-informed two- and three-dose strategies achieved $\geq 90\%$ PTA under the 93% protein-binding assumption. Sensitivity analyses showed that PTA was generally preserved at 93%–96% protein binding but declined at 97%–99%, particularly with obesity, eGFR ≥ 120 mL/min, lower-dose weekly regimens, and every-other-week dosing.

Conclusions. Body size, kidney function, and protein-binding assumptions materially influence dalbavancin PTA during prolonged therapy. Simulation-informed regimens

provide a practical outpatient framework, but higher protein-binding assumptions support cautious interpretation and consideration of individualized assessment in high-risk patients.

INTRODUCTION

Dalbavancin is a long-acting lipoglycopeptide approved for acute bacterial skin and skin-structure infections using either a single 1500-mg infusion or a two-dose regimen of 1000 mg on Day 1 and 500 mg on Day 8 [1]. While dalbavancin is approved for short-courses, it is increasingly used off-label for complex Gram-positive infections, including osteomyelitis and prosthetic joint infection, where prolonged antimicrobial exposure is required [2–4]. Clinicians therefore need practical guidance for extended outpatient dalbavancin therapy beyond the current label.

Alternative dalbavancin dosing strategies have been evaluated, including repeated weekly or biweekly infusions, weight-banded regimens, and therapeutic drug monitoring (TDM)-guided approaches [5–8]. We previously demonstrated the feasibility of individualized dalbavancin therapy within a multicenter program integrating TDM, Bayesian forecasting, and clinical pharmacologic consultation [5]. Although effective, such approaches require specialized infrastructure that is not widely available.

Population pharmacokinetic (PopPK) studies consistently identify body size and kidney function as important determinants of dalbavancin clearance [9,10]. Real-world data also suggest that patients with obesity may experience lower exposure with fixed dosing, raising concern that standard regimens may be insufficient for deep-seated Gram-positive infections requiring prolonged therapy [7, 11]. This risk may be amplified in patients with preserved or augmented kidney function due to enhanced renal elimination.

We developed a practical dalbavancin dosing nomogram incorporating body size and kidney function using a large multicenter real-world dataset and pharmacokinetic

simulations. The goal was to provide a simplified dosing strategy for clinicians managing prolonged dalbavancin therapy (generally ≥ 4 weeks) for deep-seated infections in settings where dalbavancin TDM is unavailable.

MATERIALS AND METHODS

Study Design and Patient Population

This multicenter retrospective cohort study included adult patients receiving dalbavancin for treatment of suspected or documented MRSA infections requiring prolonged therapy across five participating centers during the study period [5]. The clinical centers were Italian Infectious Diseases Departments of five Central and Northern Italy tertiary-care hospitals (Bologna, Udine, Trento, Modena and Firenze). Dalbavancin concentrations were measured before the study as part of routine clinical care. The study was approved by the local Ethics Committee of the IRCCS, Azienda Ospedaliero-Universitaria di Bologna (number 897/2021/Oss/AOUBo) and was accepted by all other centers. Patients were eligible if they received ≥ 1 dalbavancin dose and had sufficient clinical and dosing data to characterize exposure patterns. Patients receiving renal replacement therapy were excluded.

Because dalbavancin is frequently administered intermittently for treatment or suppression of chronic infections, complete dosing histories were collected for each patient including dose amount, number of administrations, and intervals between doses. Demographic and clinical data included age, sex, body weight, height, body mass index

(BMI), and comorbid conditions. Serum creatinine values obtained within seven days of dalbavancin administration were preferentially used and when multiple values were available, the measurement closest to dosing was selected. BMI categories were defined as normal weight (18.5–24.9 kg/m²), overweight (25–29.9 kg/m²), and obese (≥ 30 kg/m²). Dalbavancin plasma concentrations were quantified using a validated LC–MS/MS assay at the Clinical Pharmacology Unit (IRCCS AOU Bologna). The assay demonstrated intra-day and inter-day precision of 0.09–0.14% and 4.8–14.2%, respectively, with a lower limit of detection of 0.5 mg/L.

Kidney function was estimated using the 2021 CKD-EPI equation (with and without body surface area indexation) and the EKFC equation (with and without BSA indexation) [14,15]. Creatinine clearance was estimated using the Cockcroft–Gault equation [16]. Because the dalbavancin product label uses creatinine clearance and body weight, alternative weight descriptors including lean body weight, ideal body weight, adjusted body weight, and dosing weight were evaluated [17, 18]. A complete list of anthropometric variables and equations is provided in Table S1.

Population pharmacokinetic modeling

Previously published dalbavancin PopPK models were reviewed to define candidate structural models and parameter ranges (Table S2). Dalbavancin concentrations were analyzed using nonlinear mixed-effects modeling with the SAEM algorithm in Monolix 2024R1 (Lixoft, France).

A two-compartment model with zero-order input and first-order elimination was selected based on prior knowledge of dalbavancin disposition and model performance compared

with one- and three-compartment alternatives. Interindividual variability was assumed to follow log-normal distributions. Additive, proportional, and combined residual error models were evaluated.

Covariate analysis examined age, sex, body size metrics, and kidney function. Covariates were screened using Pearson correlations with individual random effects, and correlation between candidate covariates was also assessed to minimize redundant inclusion of highly related predictors, followed by backward elimination using the Wald test. The final model was selected based on statistical significance, biological plausibility, and model stability. Model performance was evaluated using reductions in objective function value (>3.84), Akaike information criterion, and Bayesian information criterion. Goodness-of-fit plots and relative standard errors were examined. Internal validation was performed using visual predictive checks comparing observed concentrations with simulated prediction intervals.

Protein binding assumptions

For dalbavancin, reported free fraction (f_u) estimates depend on methodology [19-21]. The product label based on regulatory review and clinical pharmacology review suggest approximately 93% plasma protein binding or a f_u of 0.07 from equilibrium dialysis studies [19]. Recent studies relying on ultracentrifugation have suggested this value to be $\leq 99\%$ ($f_u = 0.01$) in human plasma [20, 21]. Because dalbavancin is highly protein bound, small absolute changes in bound fraction produce large relative changes in estimated unbound concentrations. We therefore retained 93% protein binding as the base-case assumption to preserve comparability with historical PK-PD targets, but performed sensitivity analyses

across 93%–99% protein binding to evaluate the robustness of target attainment across pragmatic dosing regimens, and identified covariates.

Monte Carlo simulation

Monte Carlo simulations of 1000 virtual patients were performed using Simulx 2024R1. Simulations incorporated interindividual variability from the final PopPK model and were stratified by BMI and kidney function categories. Dalbavancin efficacy is associated with the fAUC/MIC ratio. Maintenance of total concentrations ≥ 8.04 mg/L corresponds to achieving a conservative PK/PD target of fAUC/MIC > 111.1 for staphylococcal pathogens with MIC ≤ 0.125 mg/L [22, 23]. Maintenance of concentrations ≥ 8.04 mg/L was therefore used as the primary probability of target attainment (PTA) criterion. Because current susceptibility breakpoints are 0.25 mg/L, simulations were also performed using a threshold of ≥ 16.08 mg/L. Simulated regimens included loading doses of 1125 mg or 1500 mg followed by potential second doses ranging from 100 mg to 1500 mg administered on Day 8. The lowest dose achieving $\geq 90\%$ PTA at 672 hours was selected as the optimal regimen for each BMI and kidney function group. Results were rounded to vial-based dosing increments consistent with clinical practice. Additional simulations evaluated three-dose regimens designed to provide 6-week and 8-week treatment coverage.

RESULTS

Study Population

A total of 311 patients were evaluated with demographic and clinical data summarized in **Table 1**. As expected, body size metrics increased progressively across

normal weight, overweight, and obese groups, while age and serum creatinine were generally similar. Notably, estimated kidney function increased with higher BMI, with a greater proportion of obese individuals classified as having $\text{eGFR} \geq 120 \text{ mL/min}$. The most common indications for dalbavancin were prosthetic joint infection (30.7%), chronic osteomyelitis (16.8%), vertebral osteomyelitis (11.9%), infective endocarditis (15.8%) and endovascular graft infections (13.9%). The initial dose of dalbavancin was 1500 mg (n=298), 1000 mg (n=11), 750 mg (n=1), or 500 mg (n=1). Most (95%) patients received two or more doses with the specific number of doses and interval between doses (after dose 2) included in Table 2. The second dose was almost exclusively within a week of the first dose, and most commonly 1500 mg (95%). The most common interval after the second dose was ≥ 672 hours (28 days) between doses, and specifically a median (IQR) of 864 (660-1152) hours.

Population pharmacokinetic model development

Dalbavancin concentrations (n=946) were measured across a broad time frame (≥ 8 weeks) allowing for robust parameter estimation (Figure 1). Exploratory analyses by BMI group identified 49% and 81% higher dalbavancin CL and peripheral volume of distribution (V2), respectively, in obese compared to normal weight patients (Table S3). Univariate analyses of the interindividual variability of CL identified both kidney function and body size as predictors. Specifically, the 2021 CKD-EPI equation and the EKFC equation without BSA indexation were the most significant eGFR models, while weight and BMI were the most significant body size parameters. Given the widespread adoption of the 2021 CKD-EPI eGFR in US electronic health systems, we selected that eGFR

equation and performed a stepwise covariate comparison that is briefly summarized in Table S4 to select the final model (Table S5). Models relying on eCL_{cr} generated higher AIC values. Figure 1 provides the VPC plot and illustrates the excellent final model fit to the data with observations available across an 8-week period after the last dose.

Simulation-based probability of target attainment

Simulations of a single dose or two-dose strategy (equivalent dose) are shown in Figure 2 based on the current product label based on the population distribution for BMI and eGFR in the current study by BMI group. As shown, likelihood of sustaining concentrations declined with increasing body size and was estimated to be the lowest, at 50% and 57% with the single and double dose strategy, respectively, in obese patients (Figure 2). The PTA after two single dose scenarios (1125 mg and 1500 mg) coupled without and with a second dose in 100 mg increments (100 mg to 1500 mg) are provided in Table S6 and Table S7. The probabilities were used to select a nomogram based on a second dose rounded to nearest 100 mg (Table S8) or 500 mg (Table S9). Similarly, the PTA for the three dose 6-week regimen (Table S10) and 8-week regimen (Table S11) were used to select optimal doses. In practical terms, patients with higher BMI and preserved kidney function were the subgroups most likely to require escalation of the second or third dalbavancin dose to maintain prolonged target attainment. Given that a recent study suggested an opportunity for 12-week regimen using three 1500 mg doses at Day 1/Day 15/Day 43 may be adequate, we tested and identified that to not be likely in obese patients with $\text{eGFR} \geq 30$ mL/min or any patients with $\text{eGFR} \geq 60$ mL/min (Table S12) [24]. We also show in Table S13 and Table S14 that use of dalbavancin trough concentrations ≥ 16.08

mg/L as a criterion represents a more conservative breakpoint-aligned exposure scenario, which may yield lower modeled probabilities of target attainment than suggested by existing real-world experience.

Taken together, these data offer a dosing by BMI group and eGFR, as well as in a pragmatic form in Table 3 (4-week regimen), Table 4 (6-week regimen), and Table 5 (8-week regimen). The PTA for the 8-week regimen were slightly below 90% in select subgroups of obese patients by BMI (Table 5) implying that TDM may be beneficial in these groups.

Protein-binding sensitivity analyses included evaluation of several alternate dosing regimens as outlined in Table S15. A heatmap of the sensitivity analyses is shown in Figure 3 with detailed values included in Table S16. These sensitivity analyses demonstrated that regimen performance is highly dependent on the assumed bound fraction. Weekly regimens, particularly 1500 mg weekly, were most robust across BMI and eGFR strata, whereas lower-dose weekly regimens and every-other-week regimens became progressively less reliable as protein binding increased. PTA was generally preserved across clinical strata at 93% to 96% protein binding, but declined at 97% to 99% binding, especially among patients with obesity, $\text{eGFR} \geq 120$ mL/min, and less intensive dosing schedules. These findings show that protein-binding uncertainty materially influences interpretation of total concentration-based PTA estimates and supports cautious interpretation of prolonged dalbavancin dosing simulations with higher protein binding expectations.

DISCUSSION

In this multicenter real-world pharmacokinetic study we developed a practical dalbavancin dosing strategy incorporating body size and kidney function that reliably maintains therapeutic exposure during prolonged therapy. Using population pharmacokinetic modeling informed by nearly one thousand measured concentrations, we demonstrate that standard fixed-dose dalbavancin regimens frequently fail to sustain target concentrations in patients with larger body size, particularly when kidney function is preserved or augmented. Model-based simulations show that modest adjustments to the second and third dose can substantially improve the probability of maintaining therapeutic concentrations across clinically relevant treatment intervals. The PK/PD thresholds used in this study should be interpreted as exposure-optimizing translational targets, rather than prospectively validated predictors of clinical success during prolonged dalbavancin therapy for deep-seated infections.

This study has several strengths. It leverages one of the largest real-world dalbavancin pharmacokinetic datasets assembled to date and includes extensive longitudinal sampling extending beyond eight weeks after dosing. The model also confirms prior pharmacokinetic observations that body size and kidney function influence dalbavancin disposition. However, BMI should be interpreted as a pragmatic body-size phenotype (and not a mechanistic marker) that improved exposure prediction and allowed construction of a clinically usable nomogram. Among kidney function estimates, the 2021 CKD-EPI equation provided a robust and clinically practical covariate for clearance, whereas Cockcroft–Gault creatinine clearance performed less well, particularly in patients with obesity. This supports use of contemporary eGFR reporting for

dalbavancin dose individualization in line with efforts to improve medication-related decision-making [25].

Protein-binding uncertainty is an important limitation and was addressed through sensitivity analysis. Because dalbavancin is highly protein bound, small absolute changes in bound fraction translate into large relative changes in estimated unbound exposure. We evaluated 93% protein binding as a base-case assumption to preserve comparability with historical regulatory PK/PD analyses, but also evaluated contemporary higher binding assumptions. Under the base-case assumption, the proposed regimens generally maintained PTA $\geq 90\%$. In contrast, PTA declined substantially at 97%–99% protein binding, especially at later time points and in patients with obesity or preserved kidney function. Recent work by Chiriac et al. further underscores this issue by demonstrating that dalbavancin PTA and cumulative fraction of response vary materially depending on the assumed protein-binding scenario, highlighting how sensitive PK/PD target interpretation can be to the selected free-fraction estimate [26].

The retrospective design precluded direct correlation between pharmacokinetic exposure and clinical outcomes. Residual confounding related to infection severity or prior antimicrobial therapy cannot be excluded. As with all retrospective multicenter datasets, inaccuracies in documentation of dosing times, dose amounts, or accompanying clinical metadata cannot be fully excluded and may have contributed to residual variability. Albumin concentrations were not consistently available across participating centers and therefore could not be robustly evaluated as a covariate. In addition, patients receiving renal replacement therapy were excluded, and the proposed nomogram should not be extrapolated to that population.

Despite these limitations, the proposed nomogram provides a practical alternative for settings where dalbavancin TDM is unavailable. TDM-guided dalbavancin dosing can support individualized prolonged therapy, but specialized assays and pharmacometric expertise are not widely available. Our simulations suggest that higher BMI and preserved or augmented kidney function identify patients most likely to need additional or more frequent dosing, and also identify as groups in whom TDM would be most valuable if available.

Importantly, clinical effectiveness studies suggest that obesity itself may not uniformly translate into worse outcomes with dalbavancin therapy. In a recent real-world analysis of invasive gram-positive infections, treatment success at 90 days was not associated with BMI despite widespread use of fixed dosing strategies [4]. These findings suggest that BMI alone may be an imperfect surrogate for exposure risk. Our results support the concept that kidney function is the dominant measurable determinant of dalbavancin exposure variability and that dose adjustments anchored to renal function may be more clinically informative than BMI alone.

Prospective studies linking pharmacokinetic exposure with clinical outcomes will be important to validate the dosing strategies proposed here. Additional research is also needed to define optimal PK/PD targets for deep-seated infections and to evaluate dosing strategies in patients with extreme obesity or rapidly changing kidney function. Future work integrating TDM with model-informed precision dosing may further refine dalbavancin therapy for complex infections.

In summary, body size and kidney function substantially influence dalbavancin pharmacokinetics, and standard fixed-dose regimens may provide inadequate exposure in patients requiring prolonged therapy. Using a large real-world dataset and pharmacokinetic simulations, we developed a practical dosing nomogram that can help clinicians personalize dalbavancin therapy during extended outpatient treatment [27]. However, protein-binding sensitivity analyses show that conclusions are strongest under the historical 93% binding assumptions, reinforcing the need for cautious interpretation, prospective validation, and individualized assessment in patients with obesity, high eGFR, or prolonged treatment durations.

Funding Sources

The authors did not receive financial support for the completion of this research or manuscript publication.

Potential Conflicts of Interest

P.V. has served as a consultant for Biomerieux, Gilead, Merck Sharp & Dohme, Pfizer, Thermo-Fisher and Venatorx, and participated in speaker bureau for Angelini, Gilead, InfectoPharm, Menarini, MerckSharp & Dohme, Nordic Pharma, and Pfizer; F.P. participated in speaker bureau for Angelini, Gilead, InfectoPharm, Menarini, Merck Sharp & Dohme, Pfizer, and Shionogi, and in advisory boards for BeiGene, Merck Sharp & Dohme, Pfizer, and Viartis. All other authors have no potential conflicts of interest to disclose with respect to the research or authorship of this article.

Author Contributions

FP and PGC were responsible for development of the initial study design, protocol, and data collection instruments. RDP curated a report to identify control patients using data from the electronic medical record. EZ, ST, SG, MM, AP and LZ performed all data collection. G. E. and J. H. conducted independent audits of data collected by other authors for internal validity. M. P. analyzed the data. M. P. wrote the initial draft of the manuscript, with critical revisions and intellectual contributions from PGC, FP and PV. All authors reviewed and approved the final version of the manuscript submitted for publication.

Declaration of generative AI use

Generative AI was not used in the manuscript preparation process.

References

1. Smith JR, Roberts KD, Rybak MJ. Dalbavancin: a novel lipoglycopeptide antibiotic with extended activity against gram-positive infections. *Infect Dis Ther*. 2015;4(3):245–258. doi:10.1007/s40121-015-0077-7.
2. Goodman-Meza D, Weiss RE, Poimboeuf ML, Feng J, Vijayan T, Martinello M, Matthews GV, Dore GJ. Comparative effectiveness of long-acting lipoglycopeptides vs standard-of-care antibiotics in serious bacterial infections. *JAMA Netw Open*. 2025;8(5):e2511641.
3. Raja K, Chen B, Chandra O, Patel M, Philips M. Budget impact and financial analysis of outpatient dalbavancin in an urban, non-teaching, community hospital. *Clin Drug Investig*. 2025;45(12):997–1005. doi:10.1007/s40261-025-01486-z.
4. Crocker J, Yunusa I, Bookstaver PB, Derrick C, Justo JA, Jones BM, White BP, Sassine J, Belk M, Pizzuti M. Real-world effectiveness of dalbavancin for invasive gram-positive bacterial infections. *Am J Health Syst Pharm*. 2025;zxaf301. doi:10.1093/ajhp/zxaf301.
5. Cojutti PG, Gatti M, Tedeschi S, Zamparini E, Meschiari M, Danzi M, Menegotto G, Cotrufo M, Soavi L, Chiari E, Ripa M, Mazzitelli M, Crapis M, Cattelan A, Parruti G, Russo A, Zammarchi L, Tascini C, Viale P, Pea F. Usefulness of a hub-and-spoke TDM-

guided expert clinical pharmacological advice program of dalbavancin for optimizing very long-term curative or suppressive treatment of chronic staphylococcal infections.

Antimicrob Agents Chemother. 2025;69(4):e01830-24. doi:10.1128/aac.01830-24.

6. Volk CF, Hutson PR, Rose WE. Simulated target attainment of multidose regimens of dalbavancin for prolonged durations of therapy. *Open Forum Infect Dis.*

2024;11(6):ofae315. doi:10.1093/ofid/ofae315.

7. Cattaneo D, Fusi M, Mariani C, Passerini M, Scandiffio L, Birindelli S, Merli S, Giorgi R, Gori A, Dolci A. Effect of body mass index on the timing of therapeutic drug monitoring-guided dalbavancin dosing in patients with osteoarticular infections. *J*

Antimicrob Chemother. 2025;80(6):1726–1732. doi:10.1093/jac/dkaf132.

8. Senneville E, Cuervo G, Gregoire M, Hidalgo-Tenorio C, Jehl F, Miro JM, Seaton A, Söderquist B, Soriano A, Thalhammer F, Pea F. Expert Opinion on Dose Regimen and Therapeutic Drug Monitoring for Long-Term Use of Dalbavancin: Expert Review Panel.

Int J Antimicrob Agents. 2023;62(5):106960. doi: 10.1016/j.ijantimicag.2023.106960.

9. Buckwalter M, Dowell JA. Population pharmacokinetic analysis of dalbavancin, a novel lipoglycopeptide. *J Clin Pharmacol.* 2005;45(11):1279–1287.

doi:10.1177/0091270005280378.

10. Carrothers TJ, Chittenden JT, Critchley I. Dalbavancin population pharmacokinetic modeling and target attainment analysis. *Clin Pharmacol Drug Dev.* 2020;9(1):21–31.

doi:10.1002/cpdd.695.

- 11.** Cattaneo D, Fusi M, Galli L, Genovese C, Giorgi R, Matone M, Merli S, Colaneri M, Gori A. Proactive therapeutic monitoring of dalbavancin concentrations in the long-term management of chronic osteoarticular and periprosthetic joint infections. *Antimicrob Agents Chemother.* 2024;68(4):e00023-24. doi:10.1128/aac.00023-24.
- 12.** Cojutti PG, Gatti M, Punt N, Douša J, Zamparini E, Tedeschi S, Viale P, Pea F. Implementation and validation of a Bayesian method for accurately forecasting duration of optimal pharmacodynamic target attainment with dalbavancin during long-term use for subacute and chronic staphylococcal infections. *Int J Antimicrob Agents.* 2024;63(1):107038. doi: 10.1016/j.ijantimicag.2023.107038
- 13.** Gatti M, Viale P, Cojutti PG, Zamparini E, De Paolis M, Giannella M, Pea F, Tedeschi S. A descriptive case series of the relationship between maintenance of conservative PK/PD efficacy thresholds of dalbavancin over time and clinical outcome in long-term treatment of staphylococcal osteoarticular infections. *Int J Antimicrob Agents.* 2023;61(5):106773. doi: 10.1016/j.ijantimicag.2023.106773
- 14.** Inker LA, Eneanya ND, Coresh J, et al. New creatinine- and cystatin C–based equations to estimate GFR without race. *N Engl J Med.* 2021;385:1737–1749.
- 15.** Pottel H, Björk J, Courbebaisse M, et al. Development and validation of a modified full age spectrum creatinine-based equation to estimate glomerular filtration rate: a cross-sectional analysis of pooled data. *Ann Intern Med.* 2021;174(2):183–191.

- 16.** Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron*. 1976;16(1):31–41.
- 17.** Pai MP. Antimicrobial dosing in specific populations and novel clinical methodologies: kidney function. *Clin Pharmacol Ther*. 2021;109(4):952–957.
- 18.** Pai MP. Antimicrobial dosing in specific populations and novel clinical methodologies: obesity. *Clin Pharmacol Ther*. 2021;109(4):942–951.
- 19.** He Y, Bergman K, Florian J. Clinical Pharmacology Review. Dalvance NDA 021-883
https://www.accessdata.fda.gov/drugsatfda_docs/nda/2014/021883Orig1s000ClinPharmR.pdf. Published 9/26/13.
- 20.** Turner NA, Xu A, Zaharoff S, Holland TL, Lodise TP. Determination of plasma protein binding of dalbavancin. *J Antimicrob Chemother*. 2022;77(7):1899-1902.
- 21.** Johnson KM, Chiodini E, Denny L, Fekry MI, Pennington C, Akers S, Rumsney J. Utilizing various plasma protein binding tools for determining the free fraction of lipophilic and lipopeptide drugs. <https://www.inotiv.com/hubfs/resources/insights-assets/Utilizing-Variou-Plasma-Protein-Binding-Tools-for-Determinating-the-Free-Fraction-of-Lipophilic-and-Lipopeptide-Drugs.pdf> . Accessed 4/7/2026.
- 22.** Lepak A, Marchillo K, VanHecker J, Andes D. Impact of Glycopeptide Resistance in *Staphylococcus aureus* on the Dalbavancin In Vivo Pharmacodynamic Target. *Antimicrob Agents Chemother*. 2015;59(12):7833-6.

23. Cojutti PG, Rinaldi M, Gatti M, Tedeschi S, Viale P, Pea F. Usefulness of therapeutic drug monitoring in estimating the duration of dalbavancin optimal target attainment in staphylococcal osteoarticular infections: a proof-of-concept. *Int J Antimicrob Agents*. 2021;58(5):106445.
24. Lin T, Senneville E, Hennart B, Valentin B, Lafon-Desmurs B, Boucher A, Pradier M, Patoz P, Robineau O, Gachet B. Real-world drug monitoring of dalbavancin using a three-dose regimen of 1500 mg at days 1, 15 and 43 in bone and joint or cardiovascular infection. *J Antimicrob Chemother*. 2026 Feb 2;81(3):dkag039.
25. St Peter WL, Bzowickyj AS, Anderson-Haag T, Awdishu L, Blackman M, Bland A, Chan E, Chmielewski C, Delgado C, Eyler R, Foster C, Hudson J, Kane-Gill SL, Kliethermes MA, Le T, Madabushi R, Martin B, Miller WG, Neumiller JJ, Philbrick AM, Roberts G, Schandorf V, Webb AJ, Wu D, Nolin TD; National Kidney Foundation Workgroup for Implementation of Race-Free eGFR-Based Medication-Related Decisions. Moving forward from Cockcroft–Gault creatinine clearance to race-free estimated glomerular filtration rate to improve medication-related decision-making in adults across healthcare settings. *Am J Health Syst Pharm*. 2025;82(12):644–659.
26. Chiriac U, Liebchen U, Frey OR, Lanzinger H, Klein S, Hoppe-Tichy T, Karck M, Meyer A, Morath B. Model-Based Dose Identification of Dalbavancin for Long-Term Suppressive Outpatient Treatment of Ventricular Assist Device Infections. *Antibiotics (Basel)*. 2024;13(11):1103.
27. Norris AH, Shrestha NK, Allison GM, Keller SC, Bhavan KP, Zurlo JJ, Hersh AL, Gorski LA, Bosso JA, Rathore MH, Arrieta A, Petrak RM, Shah A, Brown RB, Knight

SL, Umscheid CA. 2018 Infectious Diseases Society of America clinical practice guideline for the management of outpatient parenteral antimicrobial therapy. *Clin Infect Dis*. 2019;68(1):e1–e35.

Journal Pre-proof

Table 1. Population Demographics and Kidney Function by Body Mass Index Classification Group

Variables	Normal Weight (n=147)	Overweight (n=100)	Obese (n=64)	All (n=311)
<i>Continuous</i>				
Age (years)	63.6 (29%)	67.3 (20%)	61.8 (23%)	64.4 (25%)
Height (cm)	171.8 (5%)	172.1 (5%)	168.3 (5%)	171.2 (5%)
Weight (kg)	65.4 (14%)	81.4 (11%)	97.2 (15%)	77.1 (21%)
BMI (kg/m ²)	22.1 (10%)	27.4 (5%)	34.3 (14%)	26.3 (21%)
BSA (m ²)	1.80 (9%)	1.95 (8%)	2.06 (9%)	1.89 (11%)
SCr (mg/dL)	1.10 (84%)	1.04 (68%)	1.10 (62%)	1.08 (75%)
eGFR (mL/min)	74.0 (46%)	87.8 (37%)	118 (74%)	87.4 (59%)
<i>Categorical</i>				
eGFR <30	8 (5%)	2 (2%)	2 (3%)	12 (4%)
eGFR 30-59	22 (15%)	11 (11%)	10 (16%)	43 (14%)
eGFR 60-119	101 (69%)	75 (75%)	35 (55%)	211 (68%)
eCLcr ≥120	16 (11%)	12 (12%)	17 (27%)	45 (14%)
Sex (F/M)	54/93	31/69	28/36	113/198

SCr: serum creatinine; eGFR: estimated glomerular filtration rate using the CKD-EPI 2021 equation; eCLcr: estimated creatinine clearance based on the Cockcroft-Gault equation; sex: F, female, M, male. All continuous values as mean (% coefficient of variation) or number of patients if categorical

Table 2. Summary of the Number of Doses and Inter Dose Interval After the Second Dose by Body Mass Index Classification Group

Dose Number	Normal Weight	Overweight	Obese	All
1 dose	6 (4.08%)	5 (5.00%)	4 (6.25%)	15 (4.82%)
2 doses	73 (49.66%)	38 (38.00%)	36 (56.25%)	147 (47.27%)
3 doses	34 (23.13%)	31 (31.00%)	10 (15.63%)	75 (24.12%)
≥4 doses	34 (23.13%)	26 (26.00%)	14 (21.88%)	74 (23.79%)
Total Number	147	100	64	311
Inter dose Intervals				
<672 h	34 (19.9%)	35 (23.8%)	17 (20.5%)	86 (21.5%)
≥672 h	137 (80.1%)	112 (76.2%)	66 (79.5%)	315 (78.5%)
Total Intervals	171	147	83	401

Table 3. Dalbavancin by Body Mass Index Category and Kidney Function as a Single Dose or a Two-Dose Infusion (Day 1/Day 8 - 1 Week Apart) for 28 day-coverage (4-week).

eGFR (mL/min)	Dalbavancin Dose (mg)		
	Normal Weight	Overweight	Obese
<30	1125	1125	1500
30-59	1500/500	1500/500	1500/500
60-119	1500/500	1500/500	1500/1000
≥120	1500/1000	1500/1500	1500/1500

eGFR, estimated glomerular filtration rate based on the CKD-EPI 2021 equation

Table 4. Dalbavancin by Body Mass Index Category and Kidney Function as a Three-Dose Infusion (Day 1/Day 8/Day 22) for 42 day-coverage (6-weeks).

eGFR (mL/min)	Dalbavancin Dose (mg)		
	Normal Weight	Overweight	Obese
<30	1500/500/500	1500/500/500	1500/500/500
30-59	1500/500/500	1500/500/500	1500/500/500
60-119	1000/1000/1000	1000/1000/1000	1000/1000/1000
≥120	1000/1000/1000	1000/1000/1000	1500/1500/1000

eGFR, estimated glomerular filtration rate based on the CKD-EPI 2021 equation

Table 5. Dalbavancin by Body Mass Index Category and Kidney Function as a Three-Dose Infusion (Day 1/Day 15/Day 43) for 56 day-coverage (8-weeks).

eGFR (mL/min)	Dalbavancin Dose (mg)		
	Normal Weight	Overweight	Obese
<30	1500/500/500	1500/500/500	1500/500/500
30-59	1000/1000/1000	1500/1000/1000	1500/1000/1000 ^a
60-119	1500/1500/1000	1500/1500/1000	1500/1500/1000 ^b
≥120	1500/1500/1500	1500/1500/1500	1500/1500/1500 ^c

eGFR, estimated glomerular filtration rate based on the CKD-EPI 2021 equation

Probability of target attainment expected to be 79%-88%; ^aBMI ≥40 kg/m², ^bBMI ≥35 kg/m², ^cBMI ≥30 kg/m²

Figure 1. Diagnostic plot illustrating goodness-of-fit of the final model predictions (10th, 50th, 90th percentile) against the observed (scatter-plot) data illustrating an excellent fit over an 8 week (1344 hours) period after the last dose. Time in weekly intervals (168 h/week)

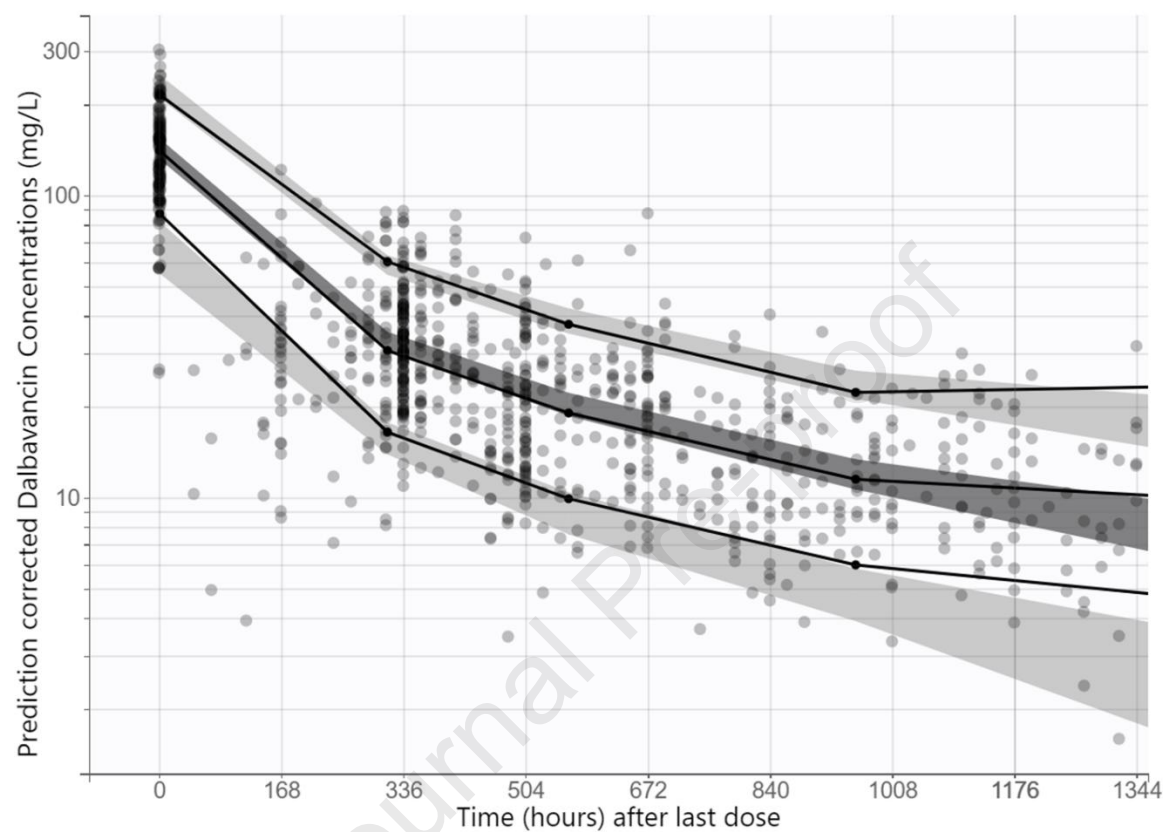


Figure 2. Median (5th, 95th percentile, light grey) and (25th, 75th percentile, dark grey) plots of the simulated profiles in the three body mass index groups with: A) Dalbavancin 1500 mg single dose infusion; B) Dalbavancin 1000 mg followed by 500 mg a week later. A visual cue at 8.04 mg/L is included (dotted line)

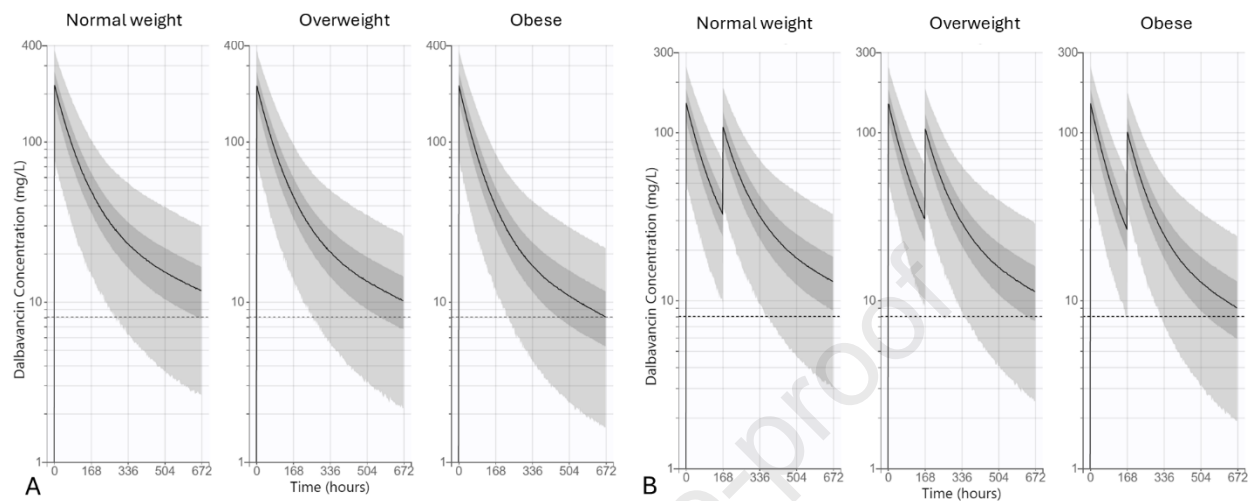
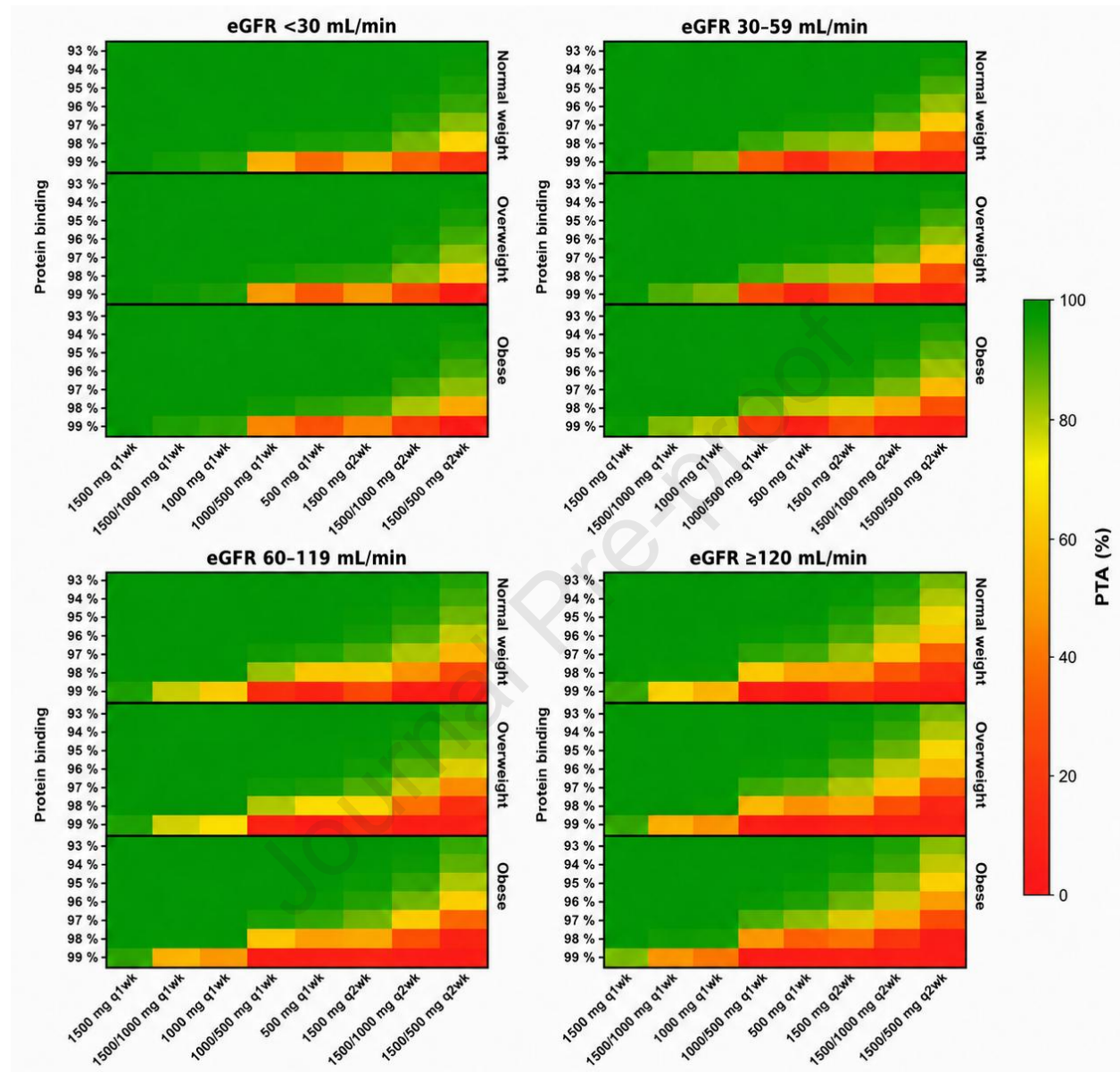


Figure 3. Protein-binding sensitivity of dalbavancin target attainment across dosing regimens, body size, and kidney function



Heatmap showing simulated probability of target attainment (PTA) across protein-binding assumptions, dosing regimens, BMI categories, and CKD-EPI eGFR groups. Each panel represents a kidney function category: eGFR <30, 30–59, 60–119, or ≥120 mL/min. Within each panel, rows are grouped by BMI category, with protein-binding assumptions ranging from 93% to 99% within each BMI group. Columns represent dalbavancin dosing regimens. Color indicates PTA, with green representing higher target attainment and red representing lower target attainment. No values are shown within cells to emphasize overall robustness patterns across protein-binding assumptions and clinical strata. PTA, probability of target attainment; eGFR, estimated glomerular filtration rate; q1wk, once weekly; q2wk, every other week.

